

Algorithms for design with CNC machines: the case study of wood furniture

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Abstract. Thanks to the application of the Industry 4.0 paradigm, contemporary factories consist of flexible production lines that can generate countless product variations without substantial increases in production costs. This study highlighted a scientific and technological gap between the flexible manufacturing system and the design system adopted to make products. In fact, commonly used CAD design technologies are static and do not allow the generation of dynamic and variable designs, causing the need to redesign models in whole or in part in order to realize variations in the generated shapes. In this paper, an algorithm-based generative design methodology oriented to the flexible manufacturing paradigm is proposed. This design approach, based on parametric modeling in Grasshopper, allows countless geometric variations of a product to be automatically generated while returning input CAD files for CNC machines. Specifically, the proposed design approach was tested by making two applications for the wood furniture industry; the output obtained in the case studies consists of a generative and parametric algorithm. The generative system provides a file for advanced manufacturing systems; in the case study, a numerically controlled laser cutter, one of the most popular machines for making flat panels from wood and metal, was chosen. The results obtained showed how the algorithmic design approach is of great importance in order to ensure customized production without substantial cost increases. This is made possible through algorithmic design automation and contemporary manufacturing technologies.

Keywords: Parametric Design · CNC · Industry 4.0.

1 Introduction

Design for manufacture and assembly (DfMA) has become an efficient design method to control the process of manufacturing and assembling. However, the existing design systems are not well suited for combining the parametric design

with DfMA. The primary purpose of DfMA method provides the ease of component prefabrication and assembly tolerance control through data model of design, modeling, fabrication and assembly. DfMA provides the potential to ensure maximum integration of knowledge from design, manufacturing and assembly. It requires customizable prefabricated elements and highly automated production systems. Current practices of designing furniture elements, and in general for wood manufacturing, often lack integrated information delivery between design, fabrication and assembly phases. In general, the integration of these processes is poorly researched in scientific papers [16, 5, 7], and the main advances are related to industrial research and are not disclosed for that reason. This paper aims to study a methodology to establish a algorithm-based generative design oriented to manufacture and assembly. The method allows the generation of parametric 3D models accompanied by information needed for manufacturing machines, such as plane representations appropriately grouped and oriented according to the characteristics of the manufacturing process in use (Laser cutting), and information for the assembly of parts. The implementation of the proposed method leads to productivity increase, to production flexibility and, since CNC machine tools are involved, to resource costs reduction. In this paper is presented an approach that aims to position itself in the scientific gap identified in the lack of integration processes between parametric and customizable CAD models and files for CNC machines.

2 Literature review

In literature, only few researchers have tried to approach this Design for manufacture and assembly automation problem as a whole over the last years. However, it is possible to find research and insights related to specific aspects of the problem itself, for example, it is possible to identify studies that have focused on reducing the geometric deviation between the designed parts and the resulting prototypes. The authors themselves investigated this problem [4, 8, 1, 9] by identifying a possible solution to geometric deviations in high-resolution SLA prints for jewelry. This geometric deviation problem is common to different categories of advanced manufacturing machines, Efendy and French [17] tried to reduce prototype deviation in CNC technologies for wood parts by acting directly on G-codes. In both cases, the results indicate that the possible variations were reduced. In addition, it is possible to identify design-related research for manufacturing at smaller scales, especially for jewelry [6, 2, 10, 3, ?]. thus, these are applications that employ high-resolution tools but generate prototypes consisting of a single or few components, so little weight is given to assembly in the parametric and generative design [11, 13, 14]. In larger-scale prototyping, however, assembly design plays a major role in the ideation and production process [19, 18, 12]. The example of prefabricated bridges [21], although far from the discussion in this paper, is emblematic in that it highlights how the design of assemblies combined with a process geared toward the manufacturability of the

object allows for substantial reductions in cost and lead time. A further point explored in depth in the literature are studies on the effects of laser cutting on panels made for the wood furniture industry. Cebrail and Tutuş [15] made comparisons between the effects on the finish of wood panels made by traditional cutting methods and by laser cutters. They identified a cutting power level of the CNC laser machine and two distinct cutting speeds, related to panel thickness, that returned an acceptable level of surface roughness of wood materials. The parameters identified in the work described, were then employed by the authors in the implementation of the algorithmic approach presented in this paper.

3 Methodology

The main purpose of the work presented is to create a system that integrates parametric modeling to manufacturing with numerically controlled machines, in the specific case study described a laser cutter is used. To this end, a design-for-manufacture and assembly approach was implemented in Grasshopper, a parametric modeling plug-in for Rhinoceros software. Parametric modeling (PD) allows an object to be designed by an explicit flow of data in the form of a graph. The dataflow permits modeling a design object as a constrained collection of schemata that are represented in the form of a graph, called Direct Acyclic Graph (DAG). DAGs can be realized by means of pre-programmed components, components produced by third parties, and by the realization of scripts in python or C #, for the realization of specific tasks, which can be combined with the grasshopper graphics language. In programming specific components, it is possible to employ the Rhinocommon library to create, manage and transform geometries, but also third-party libraries that allow temporary files to be written to the system. Specifically, such a system has been used to create files for machines containing the coordinates of the polylines describing the geometries, the speed of movement of the laser cutter head, and the cutting power level. The parameter and rule-based algorithmic design of the PD aids the implementation of the DfMA approach by allowing the unique identification of components, their positioning with respect to the global model, and then the definition of assemblies to achieve what was designed. This approach allows the creation of designs that are not only aesthetically appealing and functional, but also optimized for efficient manufacturing and assembly. Figure 1 shows the flowchart of the methodology implemented in the present work. The implemented method consists of the creation of a parametric digital model of the object to be produced, from the latter the individual components and their relative positions are identified for assembly instructions. A script extracts the polylines that will then be converted into the laser cutter's machining instructions along with the specific parameters of the machine used, such as speed and power. when a variant of the parametric model is fixed, this results in a set of G-codes, one for each component, ready for transfer to the cutting machine. Loading and unloading the machined wood panels remains the only manual operation to be performed. It is clear from the description provided that the methodology presented is generic

and can be scaled and implemented for different processing chains. In the present work, a case study was carried out for wood furniture; this choice is mainly due to the laser cutter present in the laboratory, namely a VD 1325 CO2 laser. The machine in question is an industrial laser cutter specifically for cutting organic material. The machine used has a large work surface and has a power range that allows the cutting of even thick wood panels. These features make it particularly high-performing in processing for wood furniture.

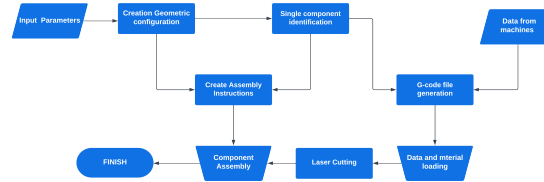


Fig. 1. Flow chart of the proposed method.

4 Results

In the present section authors report two different parametric models developed for the wood furniture sector in Grasshopper, in both case studies the proposed model was implemented. developing visual scripts combined with components programmed in python related to both the generation of the shapes themselves and the writing of the G-code. Specifically, it is therefore possible to distinguish two main steps common to both case studies: The first is related to the creation of the shapes and the identification of the components, while the second step transforms the components into lists, the geometries are then identified as discrete sets of points and polylines. The first step is strongly related to the designed form and is described in the two subsections 4.1 and 4.2. The second step consists of an algorithm in python that exploits "write()" method to create G-code files containing the discrete set of coordinates in the plane and the oriented segments joining them. This data represents the path that the machine's cutting head must take to accomplish the required machining. In the specific case, the machine is a 150-watt CO2 laser with a working plane of 2500x1350 mm. In addition to the path information the G-code must provide the power and speed of movement of the cutting head, literature analysis [15] and experimentation have shown that the optimal parameters for achieving a good finish of the cut surfaces are a constant power of 130 watts and a speed of 10 mm/s for panels with thickness ≥ 10 mm or 20 mm/s for panels with less thickness .

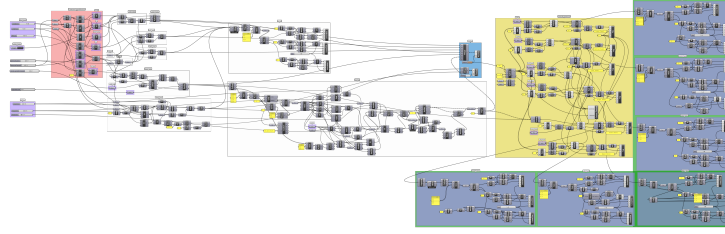


Fig. 2. DAG for the DfMA approach; the visual script generates a digital model, divided into components and G-code files for laser cutter.

4.1 Case study 1

The first case study involved the implementation of the proposed method for building a parametric kitchen module. The entire code was realized mediate the Grasshopper visual script, with some components made in python for the distribution of components on the XY plane and the component that by means of the "write()" method transforms coordinates and polylines into axis motion for the CNC machine. Figure 2 shows the visual script by means of which the first case study was implemented. as is easily deduced from the image as the complexity of the problem represented increases the complexity of the graph by means of the visual script increases and it becomes particularly difficult for external users to read. The use of groupings by color can help identify the main functions used in the visual code (Fig 2), with red highlighting the components that define anchor points, "white" highlighting the components that define the parametric shape, yellow highlighting the assembly instructions, and green highlighting the group of components that manage the distribution of plan components and the writing of the G-code. Figure 3 shows the main outputs obtained through the script made for the case study, parametric modeling combined with the DfM approach. As illustrated by Fig 3.a.b and c the visual script returns an editable digital model by changing the input parameters, with each change corresponding to a change in the components and consequently in the generatable cut file by fixing a given configuration (Fig 3.d).

4.2 Case study 2

The second case study involved the creation of a library, the script for generating the editable geometry was developed mainly by implementing a C # script inspired by fractal subdivision. While it has components for assembly and G-code generation common to the first case study, for that reason not discussed in depth in the current section. The script uses the RhinoCommon API specifically the Rhino.Geometry namespace. The first step in the process involves creating a starting rectangle that represents the overall dimensions of the library, and its center is identified. Next, a list (`List <Rectangle3d>()`) is initialized to contain

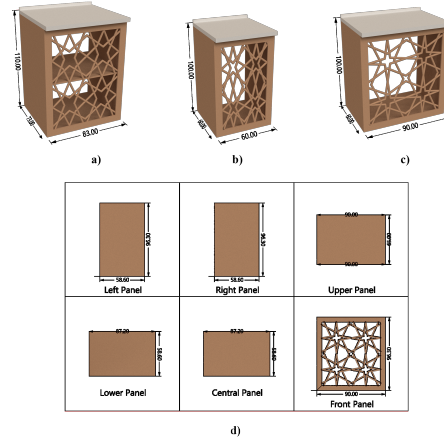


Fig. 3. Results of the first case study: the implemented method returns a parametric digital model a,b,c) and its components d) from which G-codes are extracted.

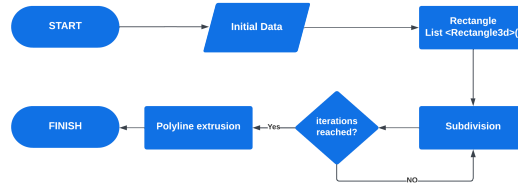


Fig. 4. Flow chart script for building a wooden bookcase using fractal subdivision.

successive subdivisions of the starting rectangle, as well as a list of the coordinates of the centers of each rectangle. After specifying two input parameters—a ratio to define the first subdivision and the desired number of iterations—a "for" loop is started that, using the principle of fractal subdivision, generates a series of rectangles within the main one. At the end of the loop, the resulting curves are extruded to create a solid. In Fig 4 a summary flowchart of the implemented script is given, the result allows design variants and files to be generated for CNC machines, some of the variations obtained are shown in Fig 5.

5 Conclusion and future works

Currently, numerically controlled technologies have greater production flexibility than design methodologies. The implementation of algorithmic and generative methodologies helps to overcome this scientific gap. The objective of this study is to present one of the potential solutions to address this problem, it allows the

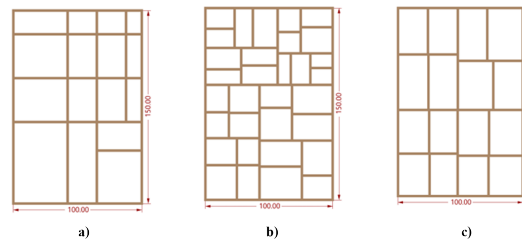


Fig. 5. Examples of parametric models obtained by providing different input parameters such as ratio and number of iterations

generation of parametric shapes oriented to manufacturing and assembly. This algorithmic approach makes it possible to go directly from the digital model to prototypes made with advanced manufacturing techniques. Future developments for the present approach involve the implementation of genetic algorithms to optimize the distribution of components to be cut by providing as input dimensions of the base panels used in order to minimize machining waste [20].

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